FORGE PREP

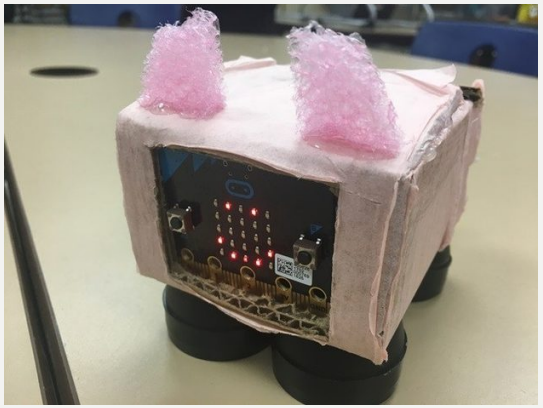
100

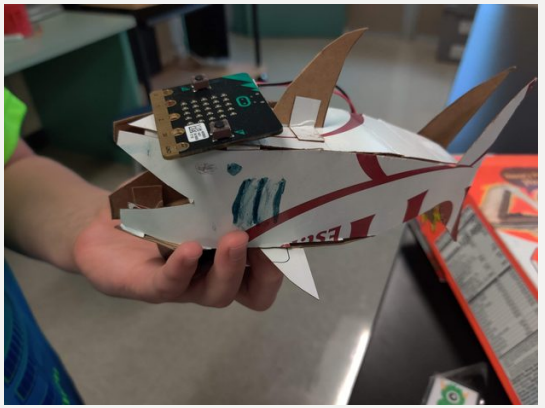
micro:pet ideas

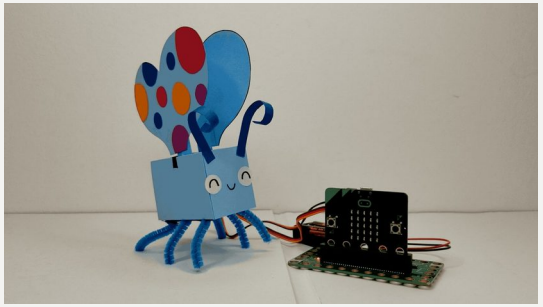
Real builds. Lots of possibilities.

Look at the body, the materials and where the micro:bit fits.

Pick a detail you like. Change it to make your own pet.







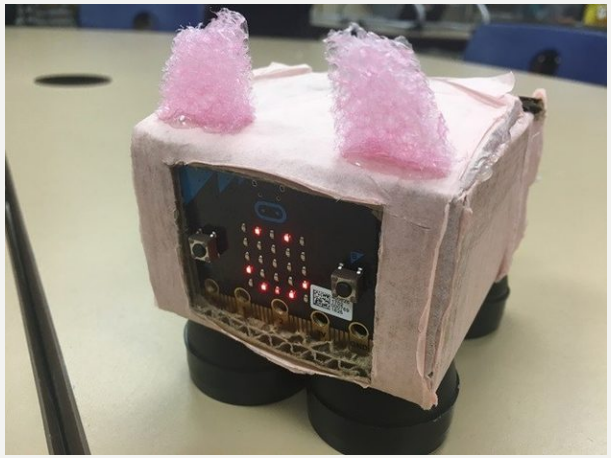


100 numbered inspiration images, including a few groups of pets.
Pictures are for ideas. Follow your class feature menu when you code.

Start with a box

FORGE PREP

One box can become lots of different pets.



01 Fuzzy-eared pig

Soft ears, a box body and four small feet.

Source: Microsoft MakeCode



02 Long-legged buddy

Folded paper legs make the body stand tall.

Source: April Smith gallery



03 Pom-pom pal

A bright body with a fluffy creature on top.

Source: April Smith gallery



04 Pink box pig

A snout, ears and a screen on the front.

Source: April Smith gallery

Start with a box

FORGE PREP

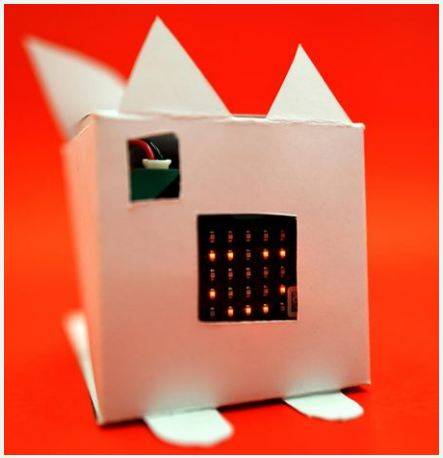
One box can become lots of different pets.



05 Pocket-console pal

A neat paper shell with tiny arms and legs.

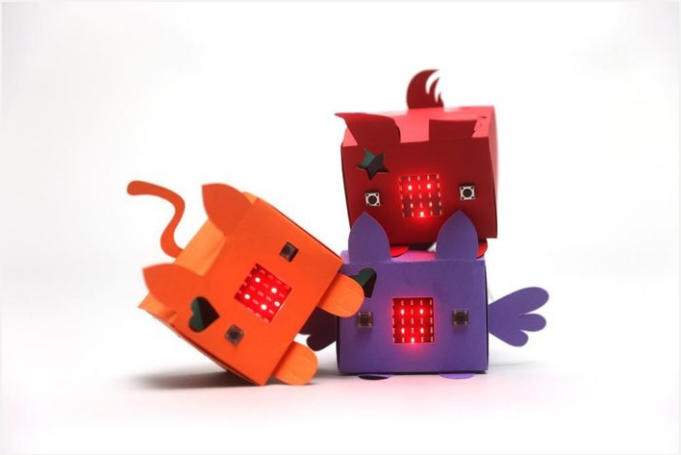
Source: April Smith gallery



06 White box cat

Two folded ears and two simple feet.

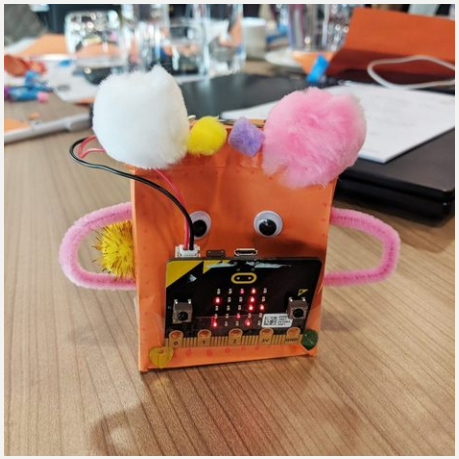
Source: City of Canada Bay



07 A whole box-pet family

Change the ears, tail and wings on the same shape.

Source: City of Canada Bay



08 Pom-pom-eared pet

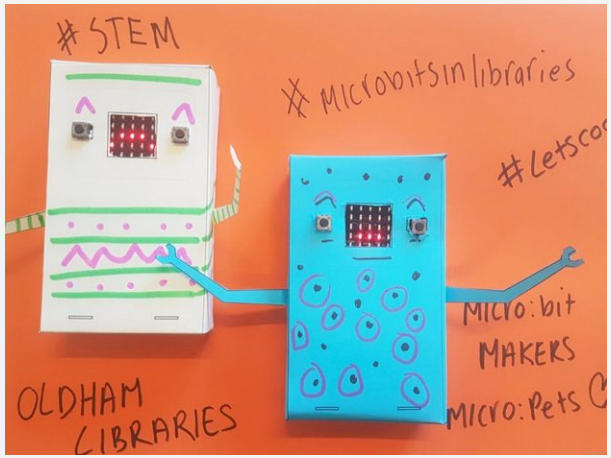
Pipe-cleaner arms and soft, colourful ears.

Source: Modern Teaching Aids

Start with a box

FORGE PREP

One box can become lots of different pets.



09 Library micro:pets

Draw a different personality on each box.

Source: April Smith gallery



10 Grey box mouse

Big ears and flat paper whiskers.

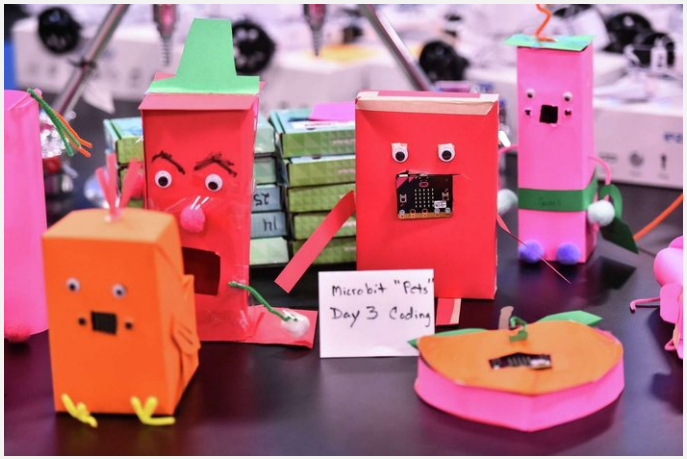
Source: Boolean Girl



11 Blue long-eared buddy

A tall box gives the ears room to stand out.

Source: Boolean Girl



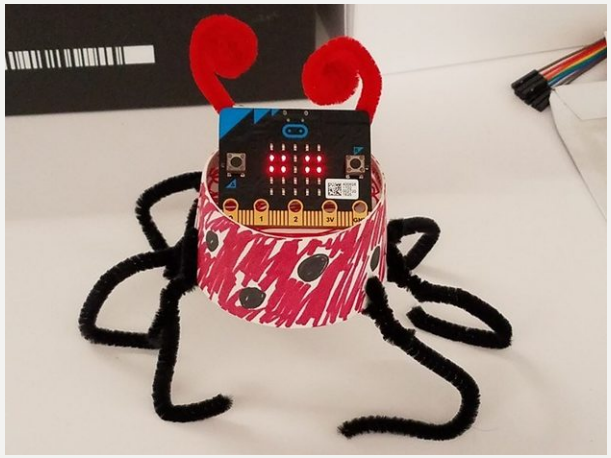
12 A table full of characters

Different sizes, colours and faces from simple boxes.

Source: San Antonio Express-News / Science Mill camp

Cups and tubes

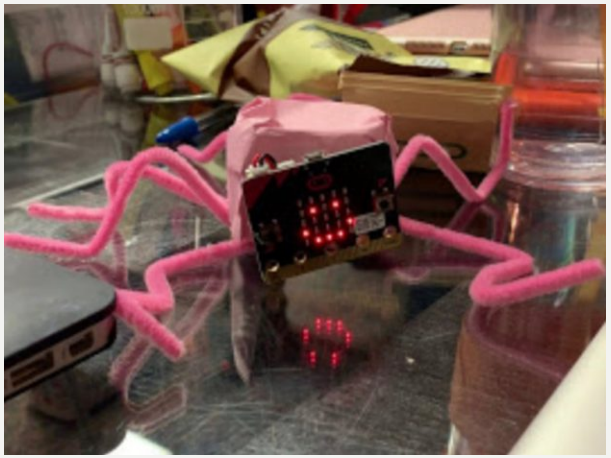
Look at what a round body makes possible.



13 Cup ladybug

A spotted cup with bendy legs and antennae.

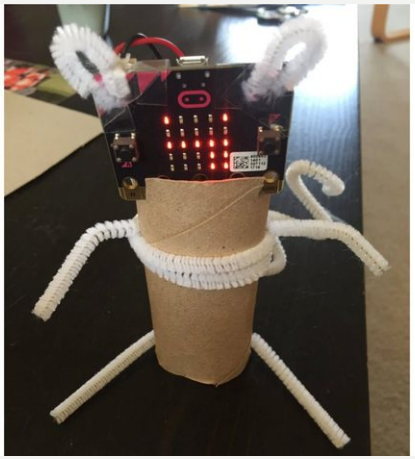
Source: Microsoft MakeCode



14 Pink-legged creature

A small body with long pipe-cleaner legs.

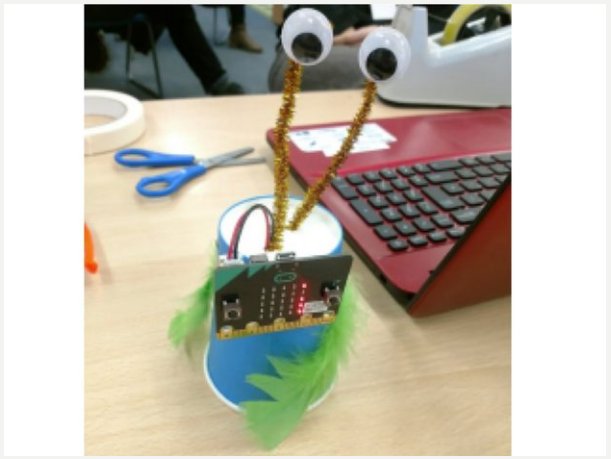
Source: April Smith gallery



15 Tube mouse

A cardboard tube, round ears and thin limbs.

Source: April Smith gallery



16 Stalk-eyed creature

Tall eyes give this tiny pet a big personality.

Source: April Smith gallery

Cups and tubes

FORGE PREP

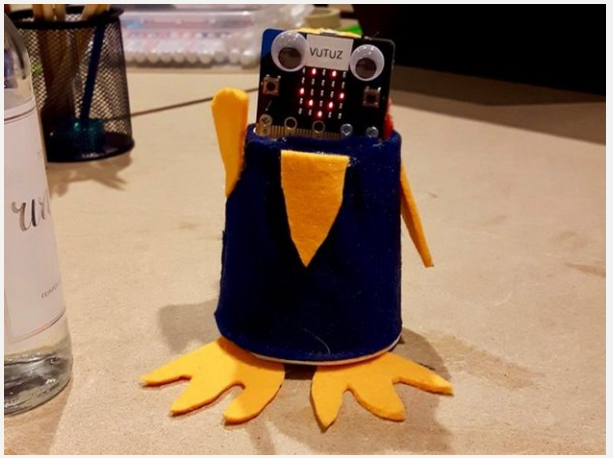
Look at what a round body makes possible.



17 Pink cup buddies

Compare the ears, eyes and feet.

Source: April Smith gallery



18 Cup penguin

A dark cup, yellow wings and wide feet.

Source: April Smith gallery

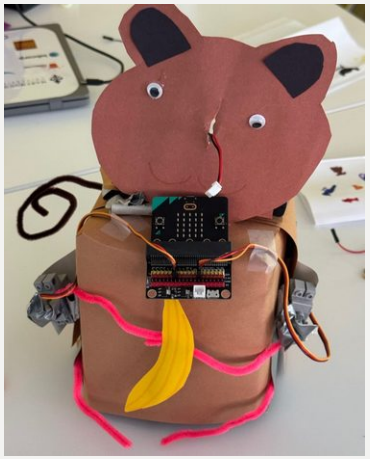


19 Tube buddy

A drawn face turns a plain tube into a character.

Source: April Smith gallery

EXTRA PARTS



20 Round-bodied mouse

A large flat head above a rounded body.

Source: Boolean Girl

Small shapes, big character

FORGE PREP

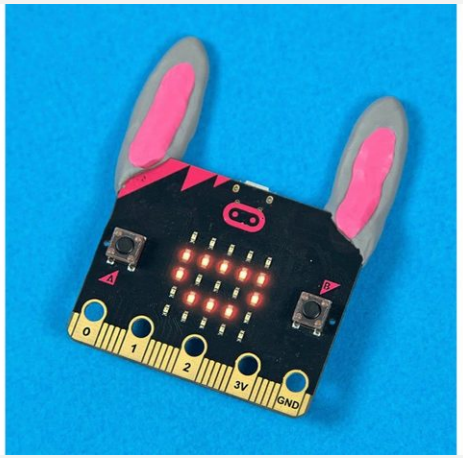
Ears, eyes and outlines can do a lot.



21 Pink face cover

A flat face with holes for the lights and buttons.

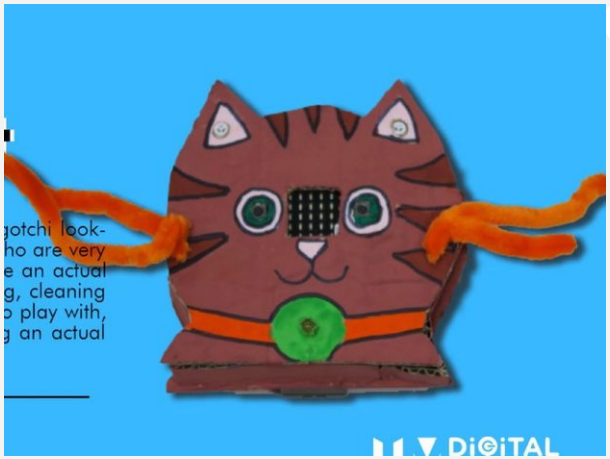
Source: April Smith gallery



22 Just add bunny ears

A simple outline around the micro:bit.

Source: April Smith gallery



23 Whiskered cat

Drawn stripes, bright whiskers and a round collar.

Source: April Smith gallery



24 Recycled-card cat

A shaped body, striped tail and LED mouth.

Source: BJC / UC Berkeley

Small shapes, big character

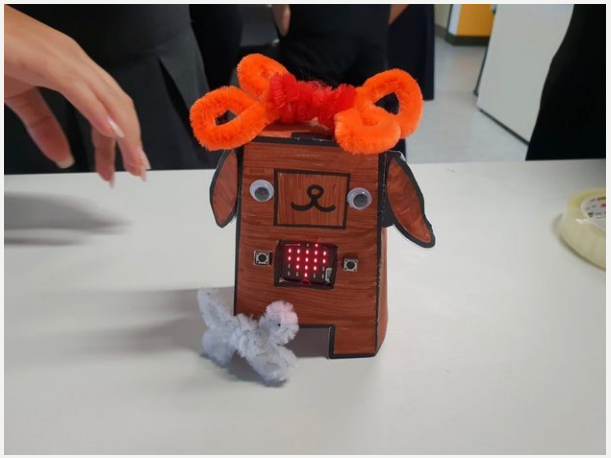
Ears, eyes and outlines can do a lot.



25 Pocket digital pet

An egg-shaped shell with a loop at the top.

Source: BJC / UC Berkeley



26 Flat dog with a bow

A drawn face and a bow make the shape recognisable.

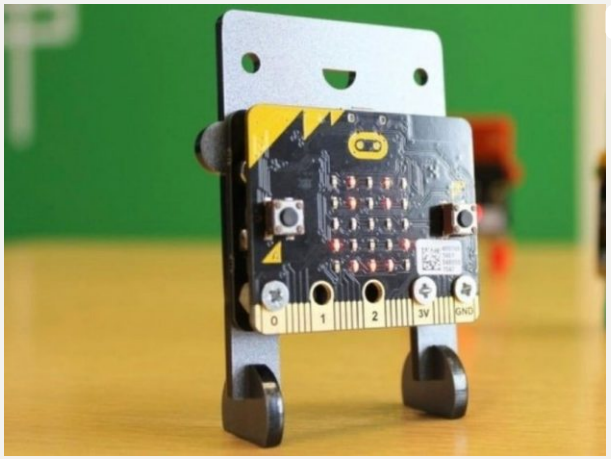
Source: April Smith gallery



27 Geometric pet family

Triangles, boxes and many-sided shapes.

Source: April Smith gallery



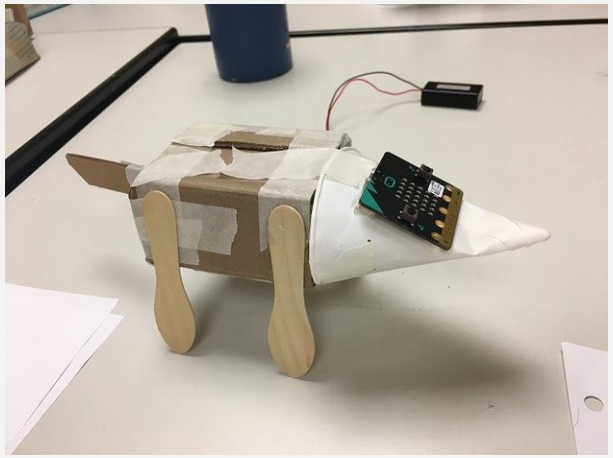
28 Little wheeled pal

A tiny upright body on a rolling base.

Source: April Smith gallery

Animal bodies

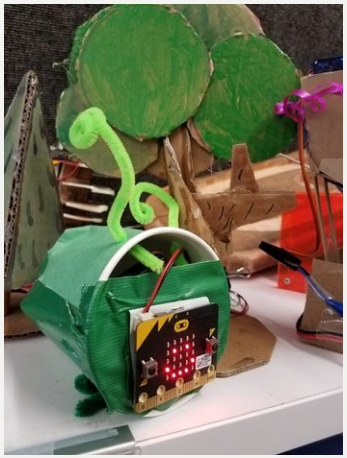
Try a different silhouette, stance or tail.



29 Craft-stick dog

A box body with flat wooden legs.

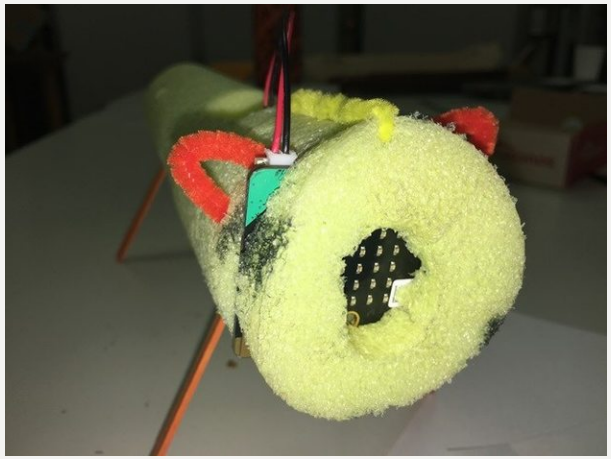
Source: Microsoft MakeCode



30 Green caterpillar

A curved green body and curled antennae.

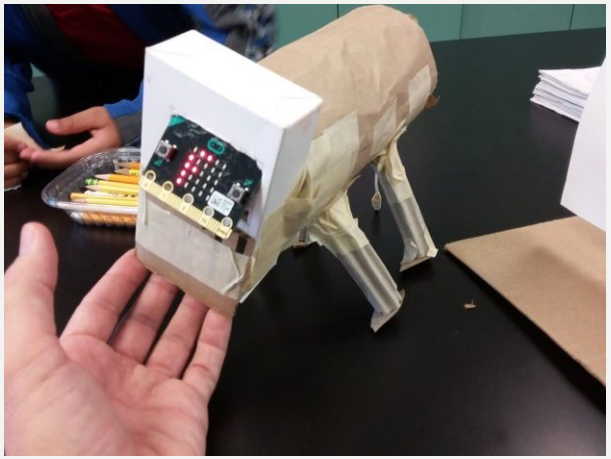
Source: Microsoft MakeCode



31 Foam fox

A rounded foam head with pointed ears.

Source: Microsoft MakeCode



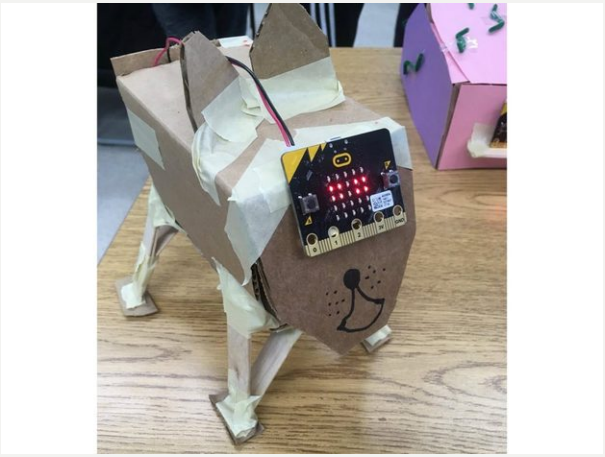
32 Long-bodied cardboard pet

A long body, four legs and a face at one end.

Source: April Smith gallery

Animal bodies

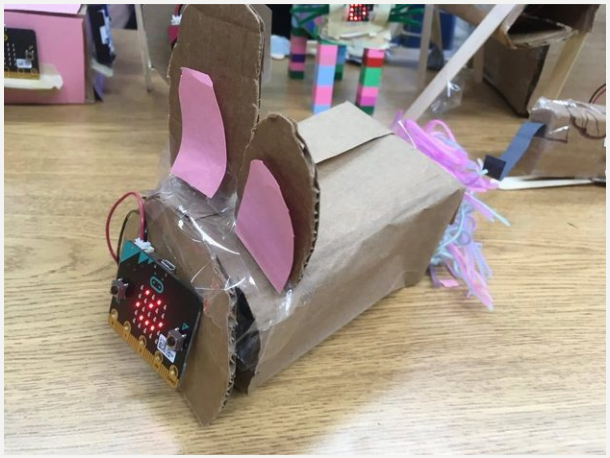
Try a different silhouette, stance or tail.



33 Long-eared cardboard pet

Tall ears and a pointed face change the silhouette.

Source: April Smith gallery



34 Cardboard pony

Notice the mane, ears and four-legged stance.

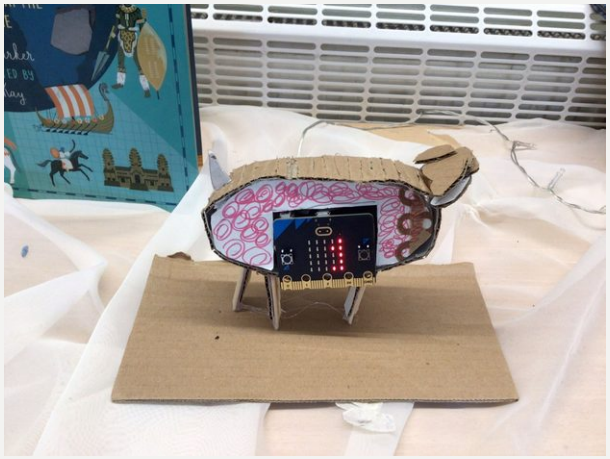
Source: April Smith gallery



35 Box bunny

A rounded face and ears on a box-shaped body.

Source: Boolean Girl



36 Cardboard animal stand

A broad body supported on a flat base.

Source: Windhill21

Mix materials

FORGE PREP

Combine shapes, textures and ways to stand.



37 Recycled-box robot

A tall package body with small feet and a wide head.

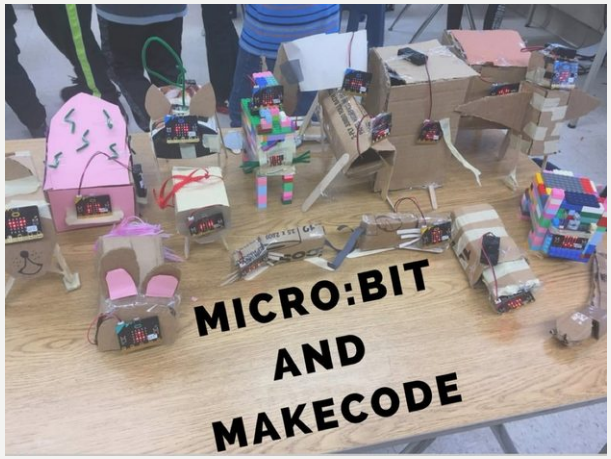
Source: Microsoft MakeCode



38 Four cardboard characters

Compare how the heads and bodies are joined.

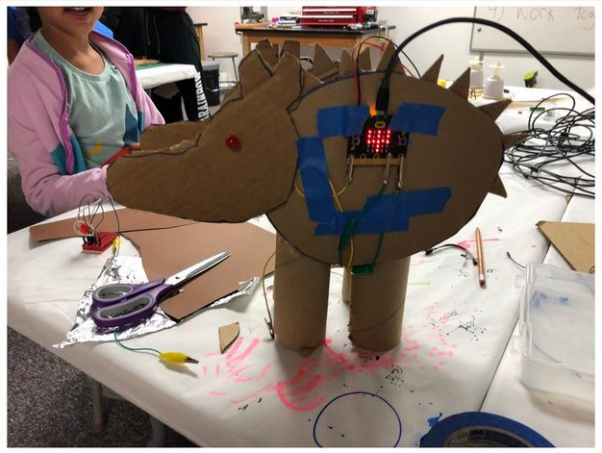
Source: April Smith gallery



39 A classroom pet collection

Lots of ways to make cardboard stand up.

Source: April Smith gallery



EXTRA PARTS

40 Spiky cardboard creature

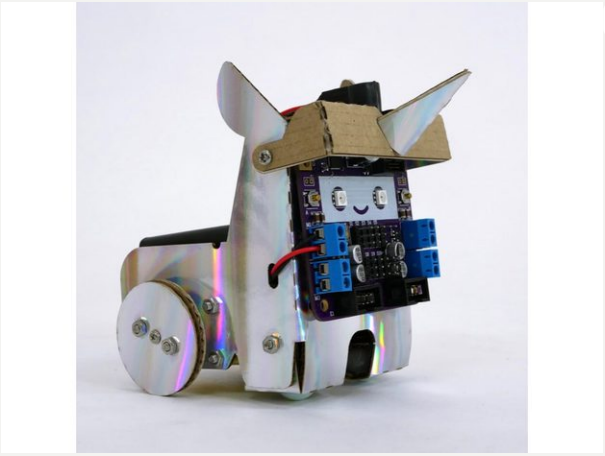
Layered cardboard creates a bold outline.

Source: April Smith gallery

Mix materials

FORGE PREP

Combine shapes, textures and ways to stand.

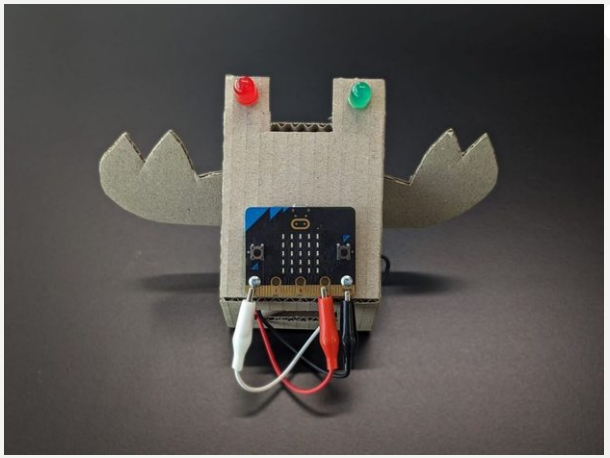


EXTRA PARTS

41 Wheeled fox

Pointed ears above a compact moving base.

Source: April Smith gallery



EXTRA PARTS

42 Antlered cardboard buddy

A flat body with large antlers and separate lights.

Source: Powerhouse



43 Soft fabric creature

A patterned fabric body with a window for the face.

Source: Primary Design Technology



EXTRA PARTS

44 Carton monsters

Reuse cartons for teeth, mouths, wings and tails.

Source: Microsoft MakeCode

Bigger ideas

FORGE PREP

Borrow a shape or detail. These builds use extra parts.



EXTRA PARTS

45 Cup-monster collection

Compare the eyes, decorations and light-up details.

Source: microbit101 / Pauline Maas

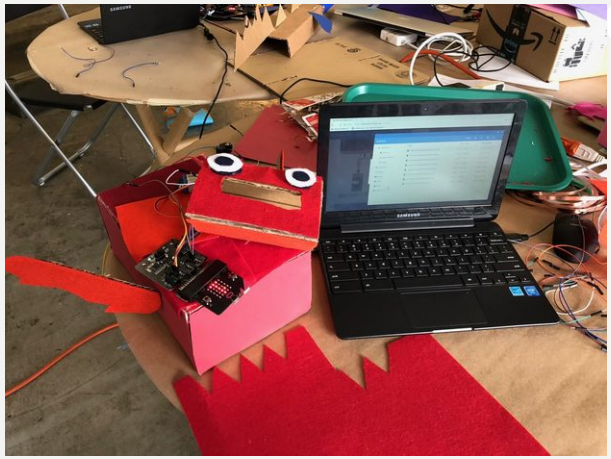


EXTRA PARTS

46 Alligator and hamster

Big eyes and bold outlines make two different characters.

Source: Wonderful Idea Co.



EXTRA PARTS

47 Wide-mouth creature

An open box becomes a huge mouth.

Source: Wonderful Idea Co.



EXTRA PARTS

48 Stacked-box creature

Stacked boxes make a tall, colourful body.

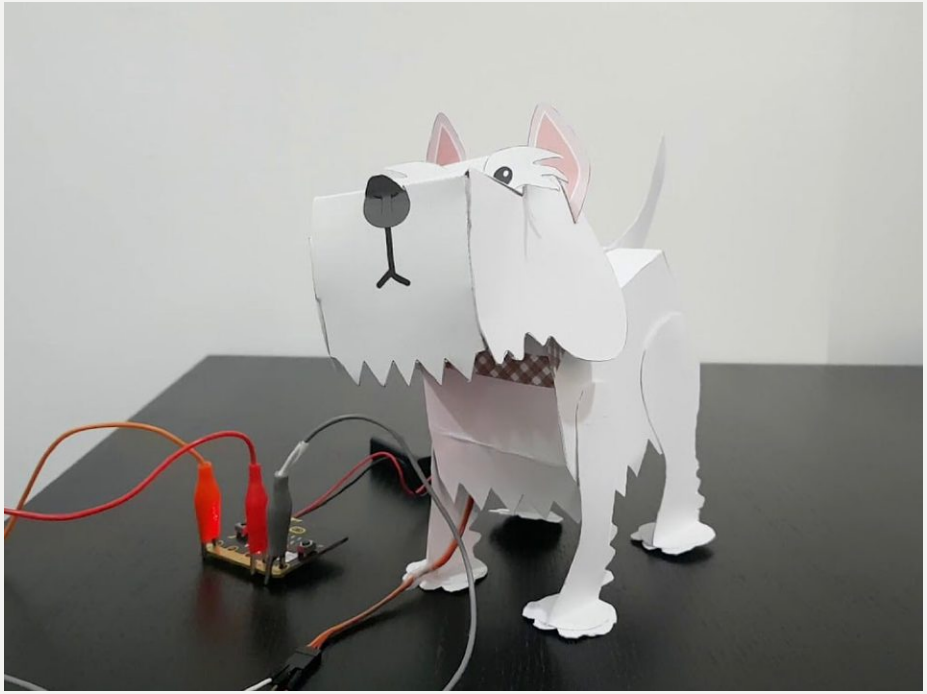
Source: Wonderful Idea Co.

Bigger ideas

FORGE PREP

Borrow a shape or detail. These builds use extra parts.

EXTRA PARTS

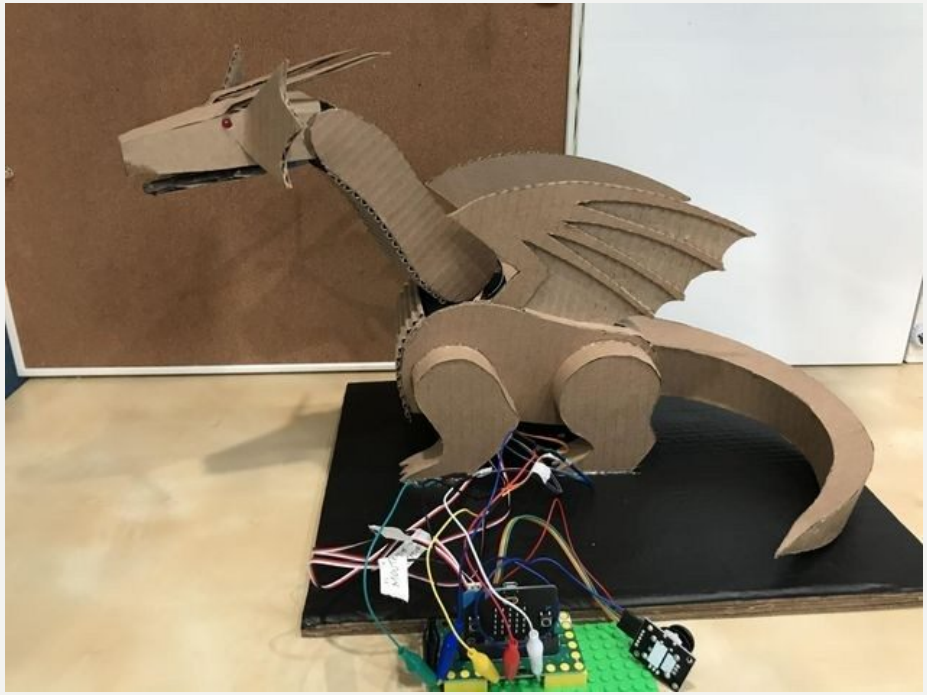


49 Paper dog

Layered paper shapes make a detailed animal body.

Source: April Smith gallery

EXTRA PARTS



50 Cardboard dragon

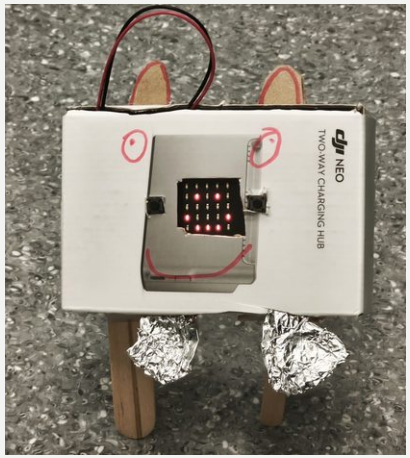
Layered wings, horns and a long curved tail.

Source: Instructables

More ways to use a box

FORGE PREP

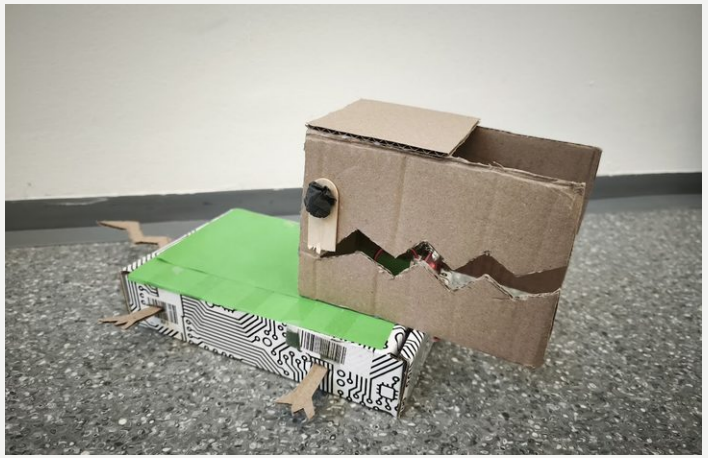
Try changing the ears, legs or shape of the body.



51 Foil-pawed friend

Shiny foil paws below a hand-drawn smile.

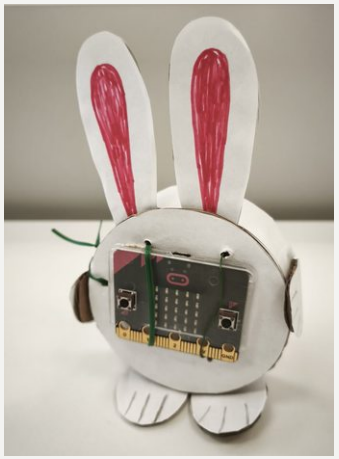
Source: Micro:bit v krouzku



52 Hinged-jaw creature

A zigzag cut turns a box into a mouth.

Source: Micro:bit v krouzku



53 Round cardboard rabbit

A round face, long ears and wide feet.

Source: Micro:bit v krouzku



54 Craft-stick cube pet

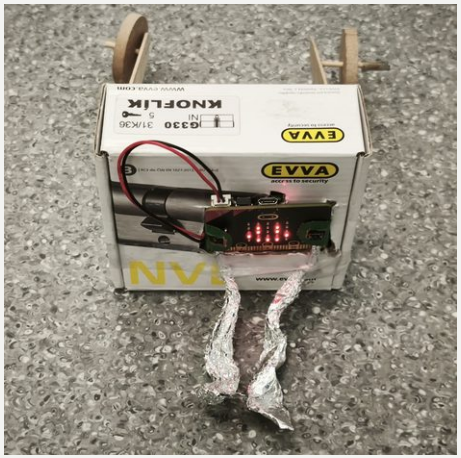
Flat wooden sticks cover the box and frame the face.

Source: Micro:bit v krouzku

More ways to use a box

FORGE PREP

Try changing the ears, legs or shape of the body.



55 Foil-legged box pet

Two shiny legs hang below a recycled package.

Source: Micro:bit v krouzku



56 Snack-box buddy

A small window makes room for the glowing face.

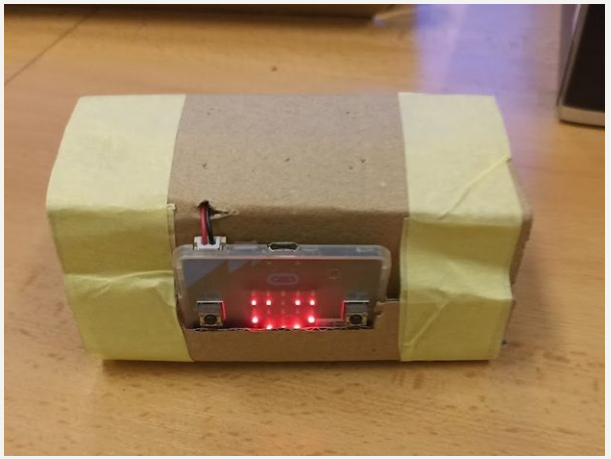
Source: Micro:bit v krouzku



57 Triangle pet

A folded triangle gives the body a different shape.

Source: Micro:bit v krouzku



58 Small taped-box pal

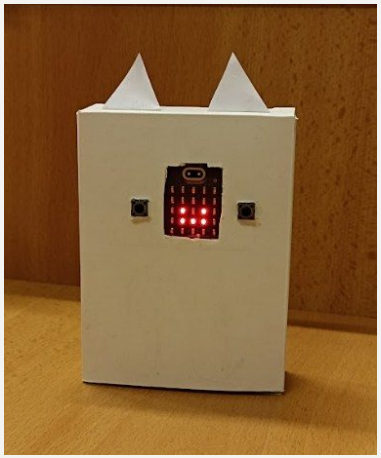
A simple box leaves the whole micro:bit easy to reach.

Source: Micro:bit v krouzku

More ways to use a box

FORGE PREP

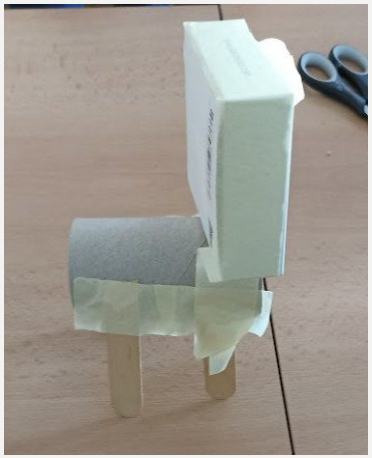
Try changing the ears, legs or shape of the body.



59 Tall white cat

Two triangular ears turn a tall box into a cat.

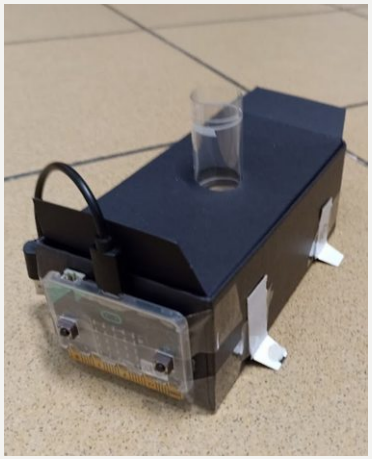
Source: Micro:bit v krouzku



60 Tube-and-box animal

A tube body, tall head and craft-stick legs.

Source: Micro:bit v krouzku



61 Long box critter

Folded paper feet support a long, low body.

Source: Micro:bit v krouzku



62 Stacked-box mouse

Round pink ears and a long tail change two boxes.

Source: Micro:bit v krouzku

Faces with personality

A few small details can make a pet recognisable.

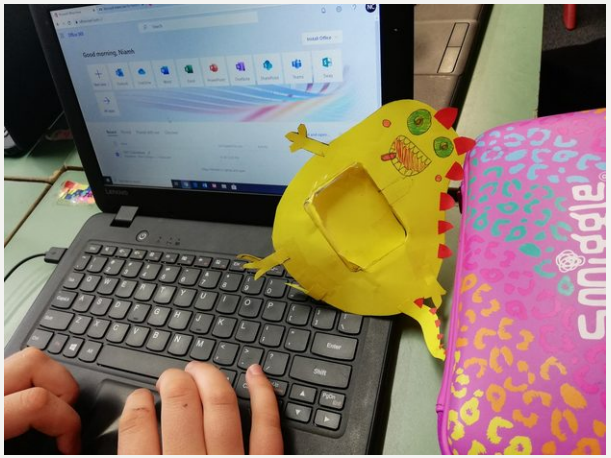
FORGE PREP



63 Yellow-eared buddy

Oversized yellow ears stand out above a small white body.

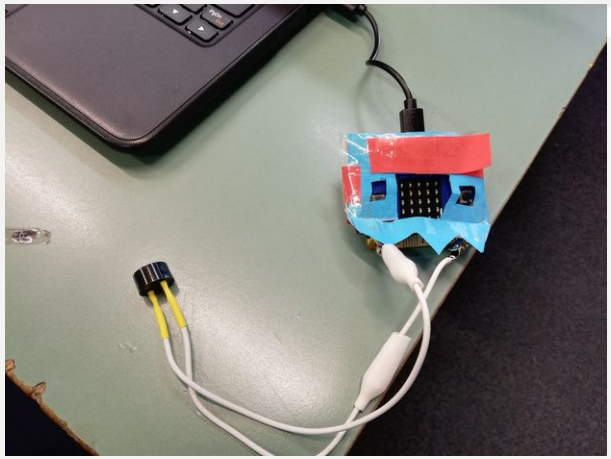
Source: Modern Teaching Aids



64 Spotted yellow creature

A round body with spots, spikes and little feet.

Source: ibpossum

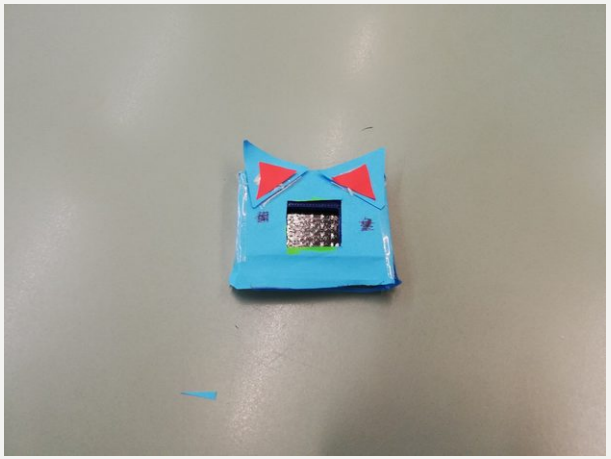


EXTRA PARTS

65 Tiny red-cap monster

A red top and zigzag edge frame the screen.

Source: ibpossum



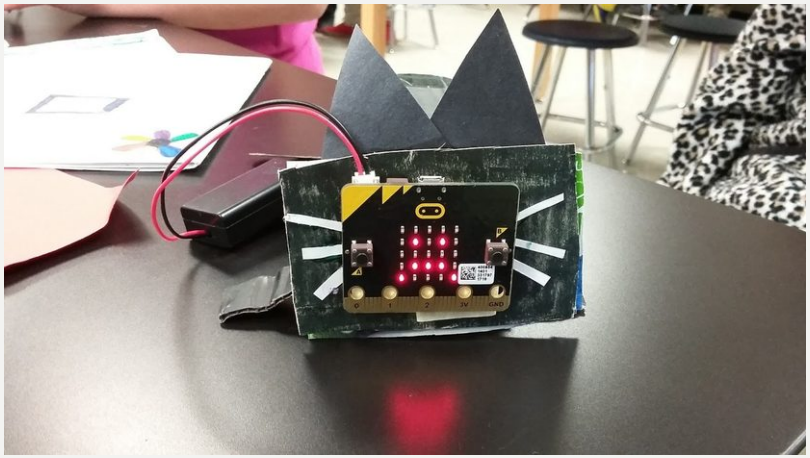
66 Blue triangle-eared pet

A folded blue body with red inner ears.

Source: ibpossum

Faces with personality

A few small details can make a pet recognisable.



67 Black cardboard cat

Pointed ears and white whiskers around an LED face.

Source: Mr. Quentin / STEM Lab



68 Pet in a tall hat

A round container becomes an oversized hat.

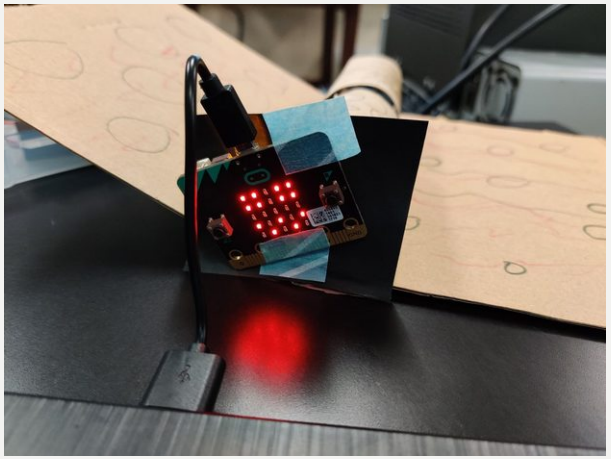
Source: Mr. Quentin / STEM Lab



69 Green box pet

Colour and texture change the look of a plain box.

Source: Mr. Quentin / STEM Lab



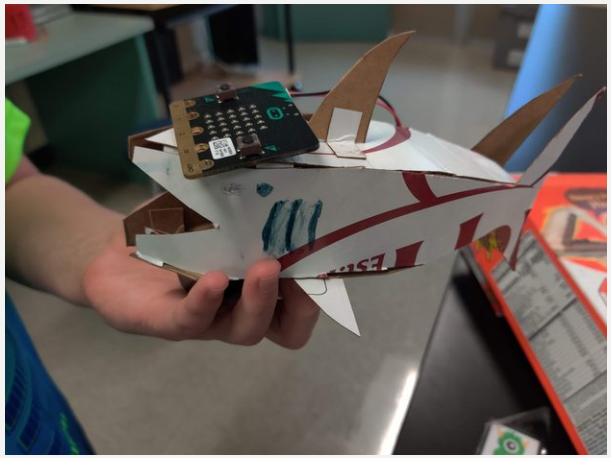
70 Wide-winged bat

Large paper wings spread out from a tiny body.

Source: Mr. Quentin / STEM Lab

Faces with personality

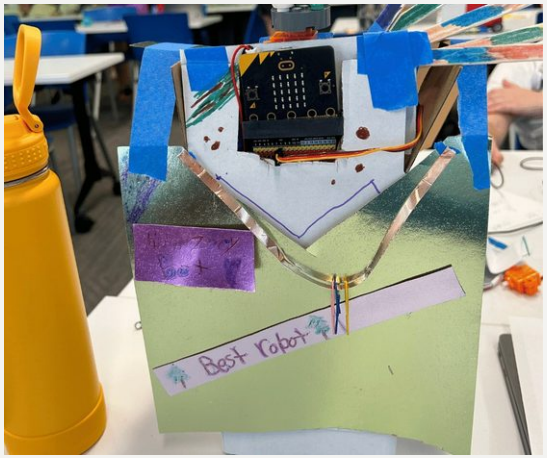
A few small details can make a pet recognisable.



71 Recycled-card shark

A pointed nose, fins and tail make a clear shark shape.

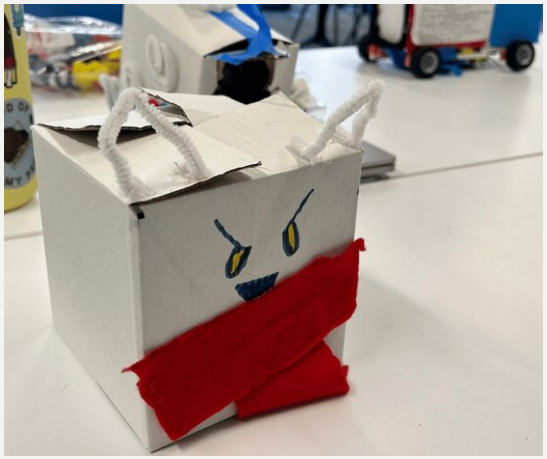
Source: Mr. Quentin / STEM Lab



72 Flat robot friend

A triangular smile and colourful patches on a flat body.

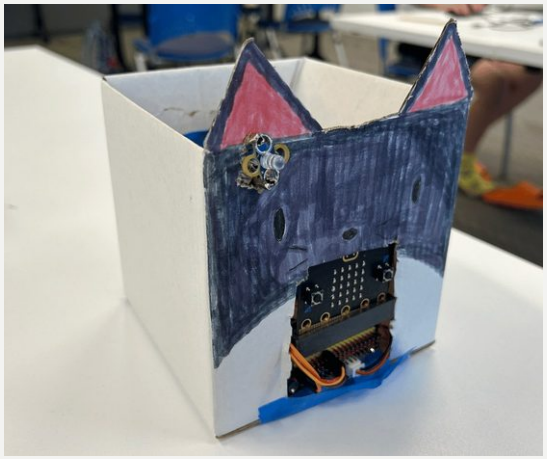
Source: Boolean Girl



73 Fierce box creature

Slanted eyes and a bold red mouth change its expression.

Source: Boolean Girl



EXTRA PARTS

74 Blue box cat

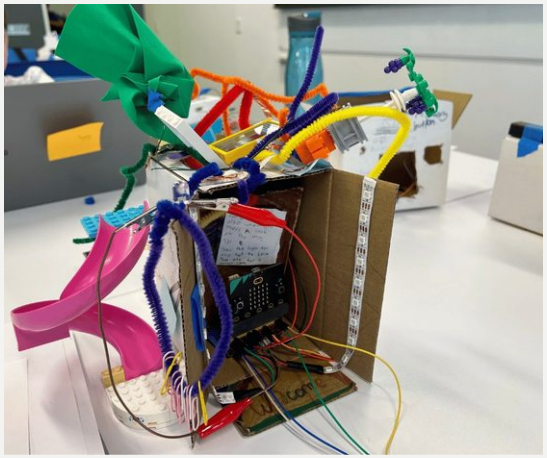
A drawn cat face wraps around a box.

Source: Boolean Girl

Colour, texture and big ears

FORGE PREP

Compare flat bodies, box bodies and soft fabric.



EXTRA PARTS

75 Wild-haired creature

Curl and bend pipe cleaners in lots of directions.

Source: Boolean Girl



76 Round fabric monster

A soft round body with bright arms and wide pink ears.

Source: Domaine des langues / RÉCIT



EXTRA PARTS

77 Flowery pet

Flower shapes and a leafy top add texture.

Source: Boolean Girl



EXTRA PARTS

78 Carrot-carrying rabbit

Long ears and a paper carrot tell you what it likes.

Source: Boolean Girl

Colour, texture and big ears

FORGE PREP

Compare flat bodies, box bodies and soft fabric.



EXTRA PARTS

79 Box panda

Black eye patches and small paws on a white box.

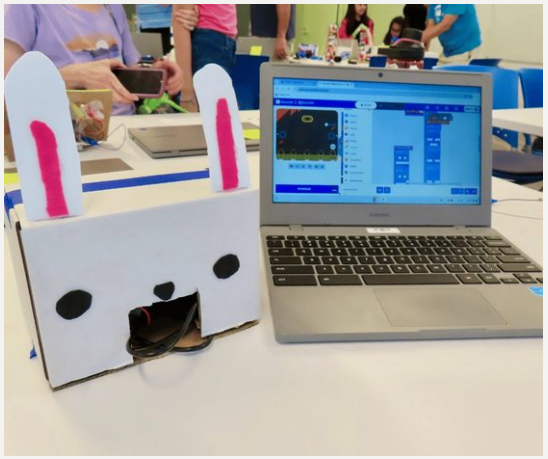
Source: Boolean Girl



80 Pink-eared pocket rabbit

A flat dark face with two long pink ears.

Source: Boolean Girl



EXTRA PARTS

81 Wide white rabbit

A wide face, pink ears and a window for the mouth.

Source: Boolean Girl



EXTRA PARTS

82 Alien family

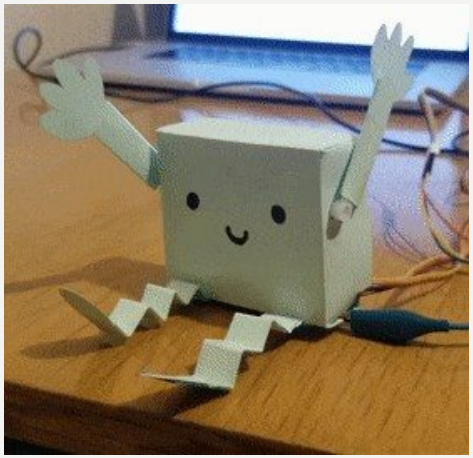
Use antennae, bright colours and different eyes on boxes.

Source: Boolean Girl

Paper creatures that move

FORGE PREP

These builds use motors or other extra parts. Borrow their shapes.

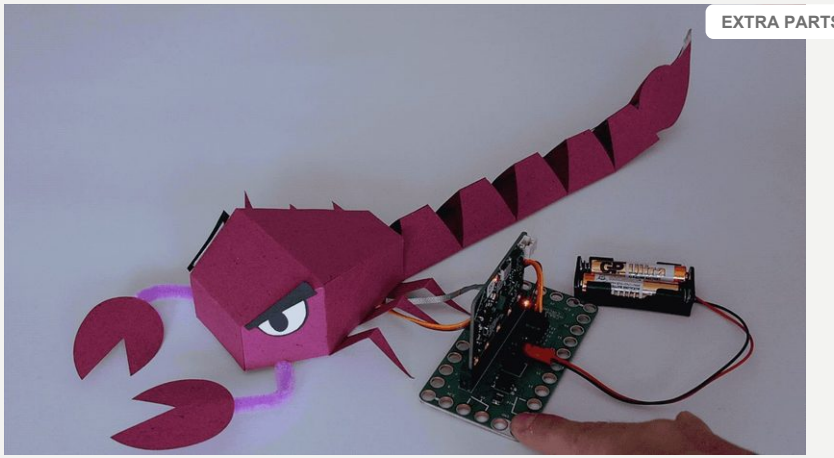


EXTRA PARTS

83 Tricklebot

Folded paper legs and two long waving arms.

Source: Jasmine Florentine

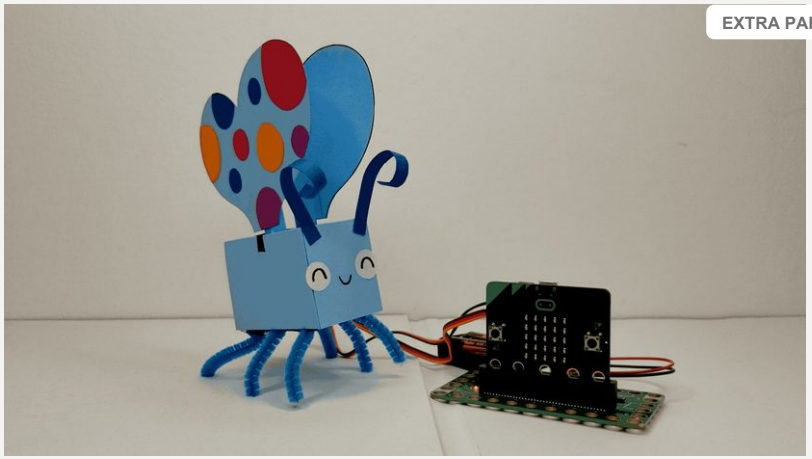


EXTRA PARTS

84 ScorpioBot

A long folded tail and two curved claws.

Source: Jasmine Florentine

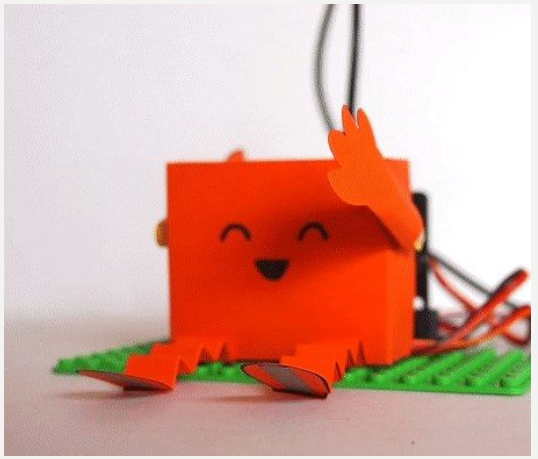


EXTRA PARTS

85 Botterfly

Big spotted wings above a tiny smiling body.

Source: Jasmine Florentine



EXTRA PARTS

86 Orange waving buddy

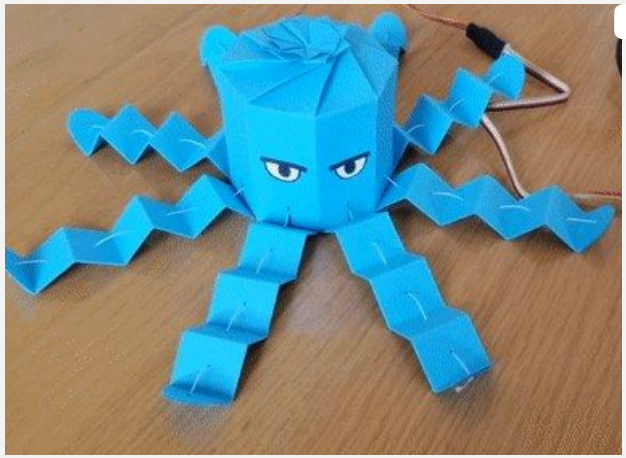
A simple box face with a big hand at the side.

Source: Jasmine Florentine

Paper creatures that move

FORGE PREP

These builds use motors or other extra parts. Borrow their shapes.



87 Octobot

Zigzag-folded arms spread out from a round body.

Source: Jasmine Florentine



88 Spotted Paper Pet

A folded body with pointed ears and a colourful pattern.

Source: Jasmine Florentine



89 Stompybot the T-Rex

A tall dinosaur with tiny arms and a long tail.

Source: Jasmine Florentine



90 Pink singing creature

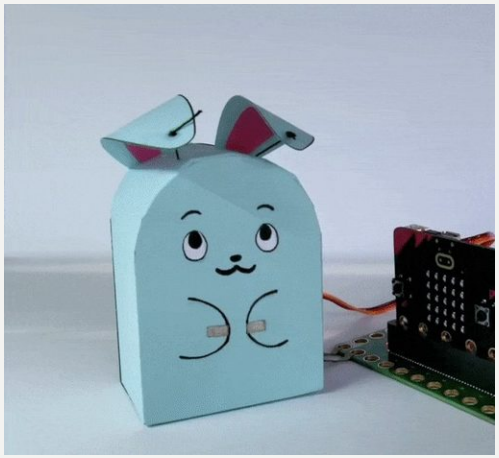
Large eyes, curved horns and a wide-open mouth.

Source: Jasmine Florentine

Paper creatures that move

FORGE PREP

These builds use motors or other extra parts. Borrow their shapes.

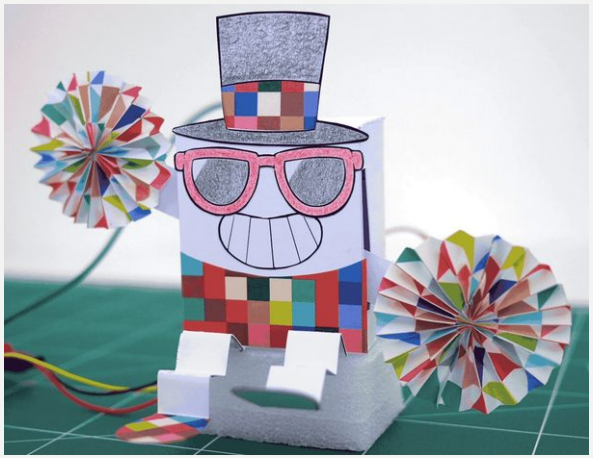


EXTRA PARTS

91 Rabbot

A rounded paper body with two floppy ears.

Source: Jasmine Florentine



EXTRA PARTS

92 Top-hat cheerbot

A tall hat, paper fans and a colourful patchwork body.

Source: Jasmine Florentine

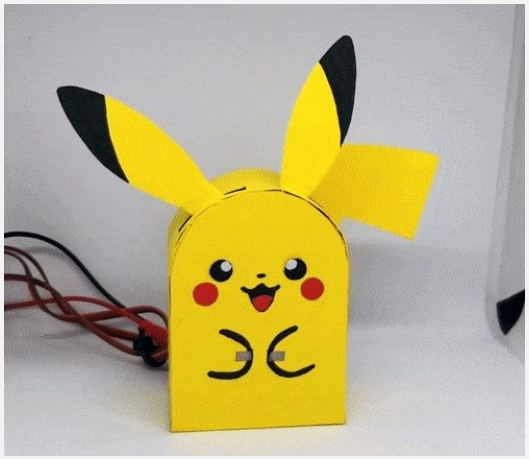


EXTRA PARTS

93 Triangle superhero

A small triangle becomes a character with a mask and cape.

Source: Jasmine Florentine



EXTRA PARTS

94 PikaBot

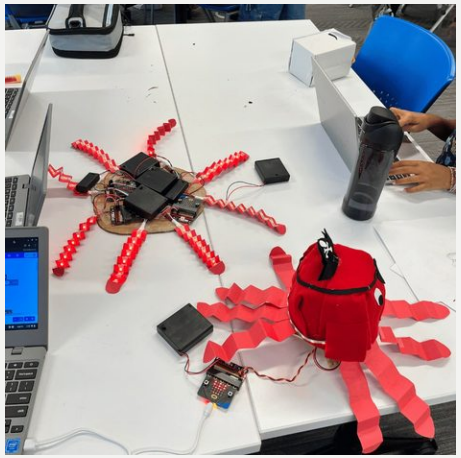
Pointed ears, red cheeks and a lightning-bolt tail.

Source: Jasmine Florentine

More big ideas to borrow

FORGE PREP

Start with one shape or detail that you like.

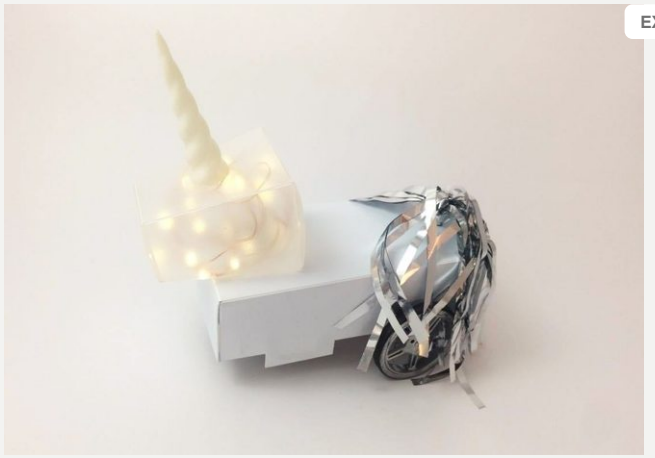


EXTRA PARTS

95 Two zigzag-armed creatures

Compare a flat body with a rounded red body.

Source: Boolean Girl

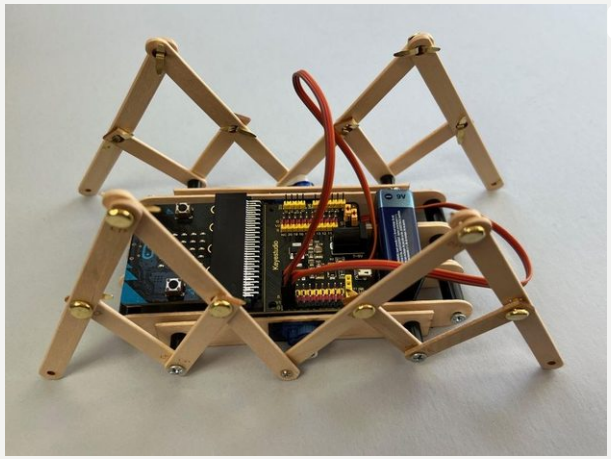


EXTRA PARTS

96 Robot unicorn

A glowing horn and shiny mane give it character.

Source: Microsoft MakeCode



EXTRA PARTS

97 Craft-stick walking bug

Long jointed legs support a small central body.

Source: hypergami / Instructables



EXTRA PARTS

98 A cardboard petting zoo

A small rabbit, a tall robot and a red box creature.

Source: U Can Too

More big ideas to borrow

FORGE PREP

Start with one shape or detail that you like.

EXTRA PARTS



99 More petting-zoo animals

Look at the owl eyes, long tongue and different ears.

Source: U Can Too

EXTRA PARTS



100 Tall blue robot friend

Stacked boxes, a big smile and a heart on its chest.

Source: micro:bit / NB Centre of Excellence

Photo sources

FORGE PREP

The image and credit on every card open its source page.

Microsoft MakeCode | Images 01, 13, 29, 30, 31, 37

makecode.microbit.org/courses/csintro/making/project

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City of Canada Bay | Images 07

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Primary Design Technology | Images 43

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Photo sources, continued

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Source links for the numbered inspiration images.

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microbit101 / Pauline Maas | Images 45

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ibpossum | Images 64, 65, 66

ibpossum.com/2019/11/13/microbit-monsters/

Mr. Quentin / STEM Lab | Images 67, 68

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Mr. Quentin / STEM Lab | Images 69, 70, 71

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www.domainelangues.qc.ca/en/resources/learning-situations-and-activities/elementary/article/monsters-at-school-project

Boolean Girl | Images 77, 78, 79, 80, 81

blog.booleangirl.org/summercamp-2024-july15

Photo sources, continued

FORGE PREP

Source links for the numbered inspiration images.

Boolean Girl | Images 82

blog.booleangirl.org/summer-camp-2025-july-21-25

Jasmine Florentine | Images 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94

www.jasmineflorentine.com/paper-robots

Boolean Girl | Images 95

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Microsoft MakeCode | Images 96

makecode.microbit.org/projects/robot-unicorn

hypergami / Instructables | Images 97

globalbyteshop.com/blogs/projects/microbit-popsicle-stickbug-walking-robot

U Can Too | Images 98, 99

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micro:bit / NB Centre of Excellence | Images 100

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Photos come from the linked project pages and galleries. “April Smith gallery” identifies the collection where those images were found; it is not a claim that April Smith photographed every example. Existing photo credits and watermarks have been preserved.

The short names and captions were written for this booklet and describe visible design details. A few pictures are still frames from the makers’ animations. Some builds use motors, extra lights, wheels or other parts: borrow their shapes and materials even if you are making a simpler pet. Sources accessed October 26, 2026.